

6.8 Leverage as geometry

The projection $\mathbf{M}$ is an $n \times n$ matrix whose entries have a direct interpretation. Since $\hat{y}_i = \sum_j m_{ij} y_j$, the entry m_{ij} is the weight observation j receives in the fitted value for observation i . So

$$m_{ij} = \frac{\partial \hat{y}_i}{\partial y_j}.$$

The diagonal entries measure how strongly each observation pulls the fitted surface towards itself.

Definition 6.8.1. Leverage

The **leverage** of observation i is $h_{ii} = m_{ii}$, the i th diagonal entry of $\mathbf{M}$. In this context $\mathbf{M}$ is often called the **hat matrix** and written $\mathbf{H}$, because it “puts the hat on” $\mathbf{y}$.

Leverage is a property of the design, not of the response. It depends on $\mathbf{y}$ not at all, and on $\mathbf{X}$ only through $\mathcal{C}(\mathbf{X})$, so it is unchanged by reparameterization.

Proposition 6.8.1. Properties of leverage

Let $\text{rank}(\mathbf{X}) = r$, and let $\mathbf{e}_i$ denote the i th standard basis vector. Then:

- (a) $h_{ii} = \|\mathbf{M}\mathbf{e}_i\|^2 = \sum_j m_{ij}^2$, the squared length of the projection of $\mathbf{e}_i$ onto $\mathcal{C}(\mathbf{X})$;
- (b) $0 \leq h_{ii} \leq 1$, and $\sum_i h_{ii} = r$, so the average leverage is r/n ;
- (c) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $h_{ii} \geq 1/n$;
- (d) $h_{ii} = 1$ iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. In that case $\hat{y}_i = y_i$ for every $\mathbf{y}$, the i th residual is always zero, and $m_{ij} = 0$ for $j \neq i$;
- (e) if $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q}$ having orthonormal columns spanning $\mathcal{C}(\mathbf{X})$, then h_{ii} is the squared length of the i th row of $\mathbf{Q}$.

Proof.

- (a) $h_{ii} = \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{M}^\top \mathbf{M} \mathbf{e}_i$, and $\mathbf{M} \mathbf{e}_i$ is the i th column of $\mathbf{M}$.
- (b) By Proposition 6.3.4(d), $0 \leq \|\mathbf{M} \mathbf{e}_i\|^2 \leq \|\mathbf{e}_i\|^2 = 1$. The sum is $\text{tr}(\mathbf{M}) = r$.
- (c) With $\mathbf{M}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$, the difference $\mathbf{M} - \mathbf{M}_0$ is a projection (Theorem 6.5.1), hence nonnegative definite, so $h_{ii} \geq (\mathbf{M}_0)_{ii} = 1/n$.
- (d) Equality holds in $\|\mathbf{M} \mathbf{e}_i\| \leq \|\mathbf{e}_i\|$ iff $(\mathbf{I} - \mathbf{M})\mathbf{e}_i = \mathbf{0}$, that is, iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. Then $\mathbf{M} \mathbf{e}_i = \mathbf{e}_i$, so column i of $\mathbf{M}$ is $\mathbf{e}_i$, and by symmetry so is row i . Hence $\hat{y}_i = y_i$.
- (e) $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$, so $m_{ii} = \sum_k q_{ik}^2$.

■

Part (d) describes the extreme case. An observation with leverage one has a coordinate direction of its own in the model space. This happens, for example, when it is the only member of a group in a one-way layout, or when a column of $\mathbf{X}$ is an indicator for that single observation. The model then fits the observation exactly, whatever its value, and the data give no check on it. High leverage is a weaker form of the same thing. If h_{ii} is close to one, then $\hat{y}_i \approx y_i$, and a gross error in y_i barely shows up in its residual.

This shows up in the residual variances. If $\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}$, then by Proposition 6.4.3(e)

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

so residuals at high-leverage points are *less* variable than others, even though those are exactly the points where an error would do the most damage. [Chapter 20](#) standardizes residuals to correct for this (Definition 20.1.1).

6.8.1 Leverage as distance from the centre

When the model has an intercept, leverage has a second description in variable space: distance from the centroid of the regressors. Write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, let $\bar{\mathbf{x}}$ be the vector of column means of $\mathbf{X}_1$, and let $\tilde{\mathbf{X}}_1 = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}_1$ be the centred regressors, assumed to have full column rank.

Proposition 6.8.2

Under these assumptions,

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where $\mathbf{x}_{(i)}$ is the i th row of $\mathbf{X}_1$, written as a column vector.

Proof. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} = n^{-1}\mathbf{1}\mathbf{1}^\top + \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1^\top$. The i th row of $\tilde{\mathbf{X}}_1$ is $(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top$. ■

The quadratic form is $(n-1)^{-1}$ times the squared Mahalanobis distance of $\mathbf{x}_{(i)}$ from the centroid, measured in the metric of the sample covariance matrix of the regressors. Leverage is therefore high for points that are unusual *jointly*, even when no single coordinate is extreme. A state with moderate poverty and moderate single parenthood, but an unusual *combination* of the two, can have high leverage. For simple regression the formula reduces to $h_{ii} = n^{-1} + (x_i - \bar{x})^2/S_{xx}$.

Example. High-leverage states

Return to the state data, now with all 51 jurisdictions including the District of Columbia, and the model of Example (Poverty and murder rates). The average leverage is $p/n = 0.078$, and the smallest is 0.021, just above the bound $1/n$. The three largest are

Jurisdiction	h_{ii}	multiple of average
District of Columbia	0.506	6.5
Mississippi	0.212	2.7
Alaska	0.177	2.3

The District of Columbia is entirely urban, with by far the highest proportion of single-parent households. It sits at a corner of the regressor cloud that no state comes near, and its leverage is about half the maximum possible. It was omitted from Example (Poverty and murder rates). Exercise (A1) asks what happens to the poverty coefficient when it is put back.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data           # District of Columbia
               kept here
X = np.column_stack([np.ones(len(data)), data["poverty"], data["single"], data["urban"]])

Q, _ = np.linalg.qr(X)                                # orthonormal basis for C(X)
h = np.sum(Q ** 2, axis=1)                             # hii = mii = ||row i of Q||2
n, p = X.shape
print("sum of leverages:", h.sum(), " (= p =", p, ")")
print("range:", h.min(), h.max(), " bounds:", 1 / n, 1)
top = np.argsort(h)[::-1][:3]
for i in top:
    print(f"{data.index[i]:22s} h = {h[i]:.3f}    ({h[i] / (p / n):.1f} x average)")
```

[Run this code online](#)

The listing computes leverages from a QR factorization rather than by forming $\mathbf{M}$. For large n the $n \times n$ matrix $\mathbf{M}$ should never be formed. [Section 6.10](#) explains why.

6.8.2 Exercises

6.8.2.1 A. Check your understanding

Exercise. A1

For simple regression, derive $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$ directly from Example (Intercept inside simple regression), and show that the largest leverage belongs to the observation with x_i farthest from $\bar{x}$.

6.8.2.2 B. Practice

Exercise. B1

Show that $m_{ij}^2 \leq m_{ii}(1 - m_{ii})$ for $i \neq j$, and hence $|m_{ij}| \leq 1/2$. When is the bound attained?

Solution. From [Proposition 6.8.1\(a\)](#), $m_{ii} = \sum_k m_{ik}^2 \geq m_{ii}^2 + m_{ij}^2$. So $m_{ij}^2 \leq m_{ii}(1 - m_{ii}) \leq 1/4$. Equality needs $m_{ii} = 1/2$ and m_{ij} the only other nonzero entry of row i . The intercept-only model with $n = 2$, where every entry of $\mathbf{M}$ is $1/2$, attains it.

Exercise. B2

Show that adding a column to $\mathbf{X}$ cannot decrease any leverage. Show that the leverages in

a one-way layout are $1/n_k$ for members of group k , and explain why an observation forming a group on its own has leverage one.

6.8.2.3 C. Going deeper

Exercise. C1

Proposition 6.8.1(d) says that $h_{ii} = 1$ iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. Here is what that means for the design. Write $\mathbf{X}_{(i)}$ for $\mathbf{X}$ with row i deleted, and let $\mathbf{D}$ be the $(n-1) \times n$ matrix that deletes the i th coordinate of a vector, so that $\mathcal{C}(\mathbf{X}_{(i)}) = \mathbf{D}\mathcal{C}(\mathbf{X})$ and $\mathcal{N}(\mathbf{D}) = \text{span}(\mathbf{e}_i)$.

- Show that $\text{rank}(\mathbf{X}_{(i)}) = \text{rank}(\mathbf{X}) - \dim(\mathcal{C}(\mathbf{X}) \cap \text{span}(\mathbf{e}_i))$, and deduce that $h_{ii} = 1$ iff deleting case i lowers the rank of the design.
- Deduce that at most $\text{rank}(\mathbf{X})$ cases can have leverage one, and that if k of them do, the remaining leverages sum to $\text{rank}(\mathbf{X}) - k$.
- Show that $\mathbf{DMD}^\top$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}_{(i)})$ when $h_{ii} = 1$, and conclude that deleting such a case changes none of the other fitted values. In what sense does the data set then contain no information about y_i ?

Solution.

- The map $\mathbf{D}$ restricted to $\mathcal{C}(\mathbf{X})$ is linear with image $\mathcal{C}(\mathbf{X}_{(i)})$ and kernel $\mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{D}) = \mathcal{C}(\mathbf{X}) \cap \text{span}(\mathbf{e}_i)$, so the rank-nullity theorem (Eq. 6.2.1) applied to that restriction gives the identity. The intersection is $\text{span}(\mathbf{e}_i)$ if $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$ and $\{\mathbf{0}\}$ otherwise, so the rank falls by one exactly when $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$, that is, when $h_{ii} = 1$.
- Leverages are nonnegative and sum to $r = \text{rank}(\mathbf{X})$ by **Proposition 6.8.1**(b), so k entries equal to one force $k \leq r$ and leave $r - k$ for the rest.
- Suppose $h_{ii} = 1$, so $\mathbf{M}\mathbf{e}_i = \mathbf{e}_i$ and, by **Proposition 6.8.1**(d), row i and column i of $\mathbf{M}$ are $\mathbf{e}_i^\top$ and $\mathbf{e}_i$. The matrix $\mathbf{DMD}^\top$ is symmetric, and since $\mathbf{D}^\top\mathbf{D} = \mathbf{I} - \mathbf{e}_i\mathbf{e}_i^\top$,

$$(\mathbf{DMD}^\top)^2 = \mathbf{DM}(\mathbf{I} - \mathbf{e}_i\mathbf{e}_i^\top)\mathbf{MD}^\top = \mathbf{D}(\mathbf{M} - \mathbf{e}_i\mathbf{e}_i^\top)\mathbf{D}^\top = \mathbf{DMD}^\top,$$

because $\mathbf{D}\mathbf{e}_i = \mathbf{0}$. Its column space is $\mathbf{D}\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X}_{(i)})$, so it is the projection of Theorem 6.3.3. The reduced fit is therefore $\mathbf{DMD}^\top\mathbf{y}_{(i)} = \mathbf{DM}(\mathbf{I} - \mathbf{e}_i\mathbf{e}_i^\top)\mathbf{y} = \mathbf{DM}\mathbf{y} - y_i\mathbf{D}\mathbf{e}_i = \mathbf{DM}\mathbf{y}$, whose j th entry is $\hat{y}_j$: the other fitted values do not move, and they never involved y_i in the first place, since $m_{ji} = 0$. The response y_i influences no fitted value but its own, which it reproduces exactly: the model has a free coordinate reserved for case i , and the data can neither check it nor borrow strength for it.

Exercise. C2

Replication bounds leverage. Suppose m cases share the same row of $\mathbf{X}$, say $\mathbf{x}_{(i)} = \mathbf{x}_{(j)}$ for all $i, j \in D$ with $|D| = m$.

- Show that rows i and j of $\mathbf{M}$ are equal for $i, j \in D$, and hence that $m_{ij} = h_{ii} = h_{jj}$ for all $i, j \in D$.
- Deduce $h_{ii} \leq 1/m$, with equality iff $m_{il} = 0$ for every $l \notin D$. Check the bound against the one-way layout of **Exercise (B2)**.
- A design point of high leverage is to be measured again. Explain what (b) says about

repeating it m times, and why this does not contradict [Exercise \(B2\)](#).

Solution.

- (a) By Theorem 6.4.1, $\mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$ for any least squares estimate, so the i th fitted value is $\mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}$. If $\mathbf{x}_{(i)} = \mathbf{x}_{(j)}$, then $\hat{y}_i = \hat{y}_j$ for every $\mathbf{y}$, which forces rows i and j of $\mathbf{M}$ to be equal. Taking the k th entry with $k \in D$ and using symmetry, $m_{ik} = m_{kk} = h_{kk}$, and also $m_{ik} = m_{ki} = h_{ii}$, so all these numbers coincide.
- (b) By [Proposition 6.8.1\(a\)](#), $h_{ii} = \sum_l m_{il}^2 \geq \sum_{l \in D} m_{il}^2 = m h_{ii}^2$, and $h_{ii} > 0$ unless the row is $\mathbf{0}$; dividing gives $h_{ii} \leq 1/m$. Equality needs the discarded terms to vanish, that is, $m_{il} = 0$ for $l \notin D$. In a one-way layout the m members of a group are exactly such a set, and the projection averages within groups, so $m_{il} = 0$ outside the group and the bound is attained: $h_{ii} = 1/m$.
- (c) Replication is the one design change that caps leverage: with m copies of a point, none of them can have leverage above $1/m$, whatever the rest of the design, so an isolated design point stops being able to dictate its own fitted value. There is no conflict with [Exercise \(B2\)](#), which adds a *column* to a fixed set of cases; here we add *cases*, and the total $\sum_i h_{ii} = \text{rank}(\mathbf{X})$ is redistributed over more of them.

Exercise. C3

Let $\mathbf{X}$ have full column rank p , write $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$, and append one more case with row $\mathbf{x}^\top$, giving $\mathbf{X}_+$ with $\mathbf{X}_+^\top \mathbf{X}_+ = \mathbf{A} + \mathbf{x}\mathbf{x}^\top$. Put $d = \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x}$.

- (a) Using Theorem 1.6.5, show that the new case has leverage $d/(1+d)$, which is always less than one.
- (b) Show that no old leverage increases: the new leverage of case i is $h_{ii} - (\mathbf{x}_{(i)}^\top \mathbf{A}^{-1} \mathbf{x})^2 / (1+d)$. When is it unchanged?
- (c) Reconcile (b) with $\sum_i h_{ii} = p$ in both designs.

Solution.

- (a) By Eq. 1.6.7, $(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{x}\mathbf{x}^\top \mathbf{A}^{-1} / (1+d)$, and $1+d > 0$ because $\mathbf{A}^{-1}$ is positive definite. So the new case has leverage $\mathbf{x}^\top (\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} \mathbf{x} = d - d^2 / (1+d) = d / (1+d) < 1$.
- (b) Applying the same formula to $\mathbf{x}_{(i)}$ gives $h_{ii}^+ = h_{ii} - (\mathbf{x}_{(i)}^\top \mathbf{A}^{-1} \mathbf{x})^2 / (1+d) \leq h_{ii}$, with equality iff $\mathbf{x}_{(i)}^\top \mathbf{A}^{-1} \mathbf{x} = 0$, that is, iff the two rows are orthogonal in the metric of $\mathbf{A}^{-1}$.
- (c) The old leverages fall by $\sum_i (\mathbf{x}_{(i)}^\top \mathbf{A}^{-1} \mathbf{x})^2 / (1+d) = \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{A} \mathbf{A}^{-1} \mathbf{x} / (1+d) = d / (1+d)$, which is exactly the leverage of the new case. The total stays at p : one more case shares the same p units of leverage.